Thor Power Board - Schematic Reference (Monitoring, eFuses, MCU)

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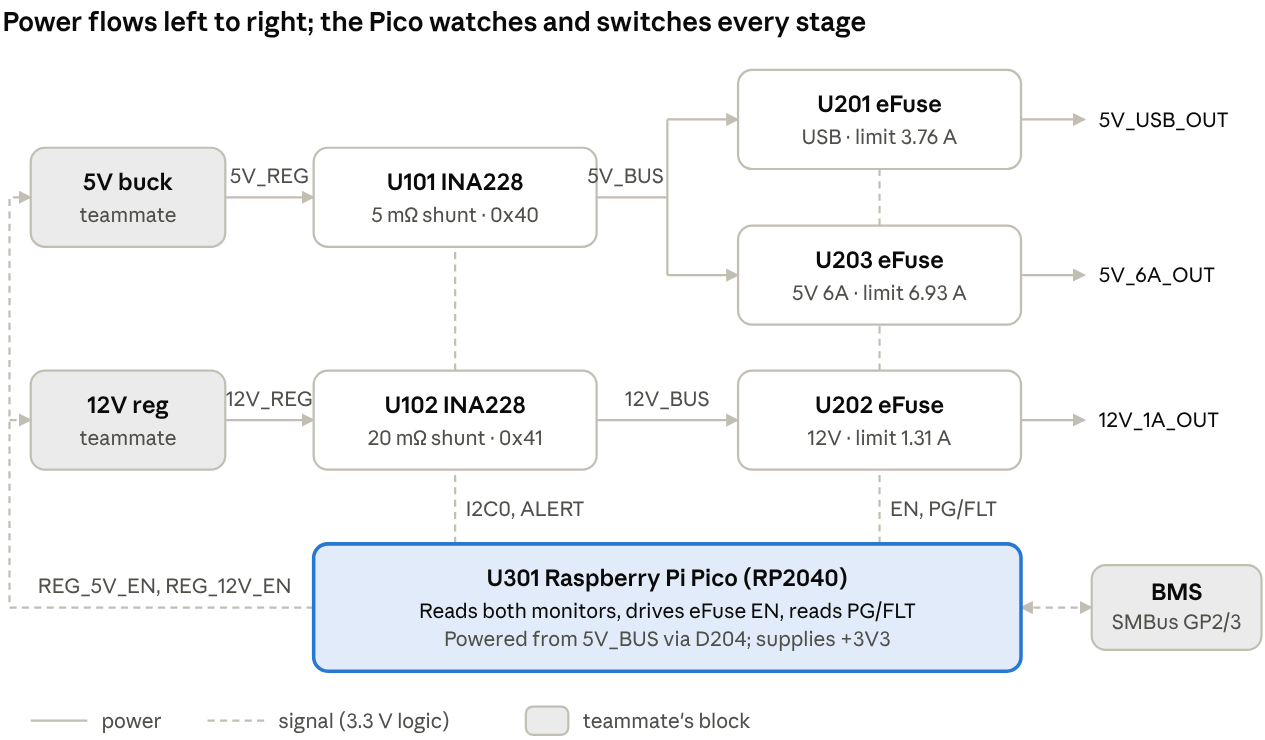
Summary

The schematic has 4 sheets: Root, Monitor, Protection, and MCU. Together they hold 2 INA228 power monitors, 3 TI eFuses, and a Raspberry Pi Pico (RP2040) that reads and controls all of them. Every one of the 192 pins was checked against the intended design, net by net (42 nets, 17 intentional no-connects), and all of them match.

One real problem turned up during verification, and it is fixed. The eFuse enable pull-ups were 100 k. The RP2040 turns on its own 50–80 k pull-down on every GPIO at reset, so EN would have sat at 1.1–1.5 V, right on the eFuse's 1.2 V on/off threshold. R204–R206 are now 10 k, which puts EN at 2.75 V or higher. The KiCad zip, PDF and BOM are all updated.

This sheet connects to the rest of Thor through 11 net names (see *Interface nets for merging*). Those names have to match the teammates' sheets exactly when the schematics are joined.

Architecture



Thor power path · 2 monitors, 3 eFuses, 1 MCU

Shaded boxes belong to teammates. Everything else is on these three sheets, and the net names on the arrows are what joins them.

Interface nets for merging

KiCad joins sheets by **name**. A global label or power symbol with the same spelling on two sheets becomes one wire. These are the nets that cross into teammates' blocks:

Net	Direction (from this sheet)	Comes from / goes to	Lands on here	Electrical
5V_REG	In	5V buck regulator output	RS101 pin 1, R101 (U101 IN+)	5 V, up to ~10 A total
12V_REG	In	12V regulator output	RS102 pin 1, R103 (U102 IN+)	12 V, up to ~1.3 A

Net	Direction (from this sheet)	Comes from / goes to	Lands on here	Electrical
5V_USB_OUT	Out	USB output connector	U201 OUT	5 V, limit 3.76 A
5V_6A_OUT	Out	5V 6A output connector	U203 OUT	5 V, limit 6.93 A
12V_1A_OUT	Out	12V 1A output connector	U202 OUT	12 V, limit 1.31 A
REG_5V_EN	Out	5V buck EN pin	Pico GP14 (pin 19)	3.3 V logic, push-pull
REG_12V_EN	Out	12V regulator EN pin	Pico GP15 (pin 20)	3.3 V logic, push-pull
BMS_SDA	Bidir	BMS SMBus data	Pico GP2 (pin 4)	Open drain; pull-ups go on the BMS sheet, to 3.3 V only
BMS_SCL	Out	BMS SMBus clock	Pico GP3 (pin 5)	Same as above
GND	Power	Every sheet	All grounds	Single ground symbol name GND
+3V3	Power	**Only** the Pico's 3V3(OUT)	INA228s, pull-ups	300 mA max from the Pico

These nets are also global but stay inside this design: 5V_BUS, 12V_BUS, I2C_SDA/SCL, ALERT_5V/12V, EFUSE_*_EN, PG_USB, PG_12V, FLT_6A. A teammate must not reuse any of those names for a different signal.

Two merge traps to watch:

- **+3V3 is a power symbol, so it is global.** If another sheet has its own 3.3 V regulator also named +3V3, KiCad will short it to the Pico's 3V3(OUT). Either share one source on purpose or rename one of them (for example +3V3_PICO).
- **The regulators' EN inputs must accept 3.3 V logic as "on".** The RP2040 cannot drive 5 V. If a regulator's EN needs a higher voltage, add a level shifter or a transistor on that sheet.

Monitoring sheet (page 2)

Two INA228s share one I2C bus (I2C0 on the Pico). Each one measures the voltage across a shunt resistor (current) and the voltage of the rail itself (bus voltage).

Item	CH1: U101	CH2: U102
What it measures	Whole 5V bus, **before** the split into USB and 6A outputs	12V rail, before the 12V eFuse
Shunt	RS101 5 mΩ 1% 2512, WSL25125L000FEA (5V_REG → 5V_BUS)	RS102 20 mΩ 1% 1206, WSL1206R0200FEA (12V_REG → 12V_BUS)
Shunt voltage at full load	9 A × 5 mΩ = 45 mV (range ±163.84 mV)	1 A × 20 mΩ = 20 mV
Shunt heat at full load	$9^2 \times 0.005 = 0.41$ W (rated 1 W)	$1^2 \times 0.02 = 0.02$ W (rated 0.25 W)
IN+ / IN- series resistors	R101 / R102, 10 Ω	R103 / R104, 10 Ω
Filter cap across IN+/IN-	C103 100 nF, DNP (footprint only)	C104 100 nF, DNP
VBUS pin	5V_BUS	12V_BUS
VS supply + decoupling	+3V3, C101 100 nF	+3V3, C102 100 nF
Address pins	A1 = GND, A0 = GND → **0x40**	A1 = GND, A0 = VS → **0x41**
ALERT (open drain, active low)	ALERT_5V, R105 10 k to +3V3 → GP6	ALERT_12V, R106 10 k to +3V3 → GP7

Shared parts: R107 and R108 (4.7 k) pull I2C_SDA and I2C_SCL up to +3V3, one pair for the whole bus. C105 (1 µF) is bulk capacitance on +3V3 near the monitors.

Firmware register values. CURRENT_LSB is chosen as 12 A / 2^19 for CH1 and 1.5 A / 2^19 for CH2. SHUNT_CAL = 13107.2 × 10^6 × CURRENT_LSB × R_SHUNT.

Register	CH1 (0x40)	CH2 (0x41)	Meaning
CONFIG	0x0000	0x0000	±163.84 mV shunt range
ADC_CONFIG	0xFB6B	0xFB6B	Continuous bus, shunt and temperature, with averaging
SHUNT_CAL	1500	750	Full scale 12 A / 1.5 A
DIAG_ALRT	0x8001	0x8001	Latched alert, conversion-ready flag
SOVL	10000	4800	Over-current alert at 10 A / 1.2 A (5 µV per bit)
BUVL	1520	3648	Under-voltage alert at 4.75 V / 11.4 V (3.125 mV per bit)

The earlier monitoring doc lists the 5 mΩ shunt as WSL2512R0050FEA. That is the wrong part number. The schematic and BOM use the correct one, WSL25125L000FEA.

Protection sheet (page 3)

Each output has its own eFuse. When a load shorts or pulls too much current, the eFuse shuts off its output and retries on its own about every 100 ms, so nothing has to be replaced. The MCU can also turn each output off or reset it through its EN pin.

Setting	U201: 5V USB 3A	U202: 12V 1A	U203: 5V 6A
Part	TPS259474ARPWR (QFN-10)	TPS259474ARPWR (QFN-10)	TPS259480AYWPR (WCSP-12)
Input → output	5V_BUS → 5V_USB_OUT	12V_BUS → 12V_1A_OUT	5V_BUS → 5V_6A_OUT
Overload behavior	Circuit breaker, auto-retry 110 ms	Circuit breaker, auto-retry 110 ms	Active current limit, auto-retry 103 ms
Current-limit resistor (ILM)	R201 887 Ω → 3334 / 887 = **3.76 A**	R202 2.55 k → 3334 / 2550 = **1.31 A**	R203 698 Ω → 4834 / 698 = **6.93 A**
ITIMER cap	C201 2.2 nF C0G	C202 2.2 nF C0G	C203 2.2 nF C0G

Setting	U201: 5V USB 3A	U202: 12V 1A	U203: 5V 6A
dVdt (soft-start) cap	C204 2.2 nF → 0.91 V/ms, about 5.5 ms to 5 V	C205 2.2 nF → 0.91 V/ms, about 13 ms to 12 V	C206 4.7 nF → 1.06 V/ms, about 4.7 ms to 5 V
Over-voltage cutoff (OVLO)	R207 100 k / R210 24.9 k → **6.02 V**	R208 100 k / R211 9.31 k → **14.1 V**	R209 100 k / R212 24.9 k → **6.02 V**
Status output	PG → PG_USB; high when OUT > 4.44 V (R213 47 k / R214 17.4 k)	PG → PG_12V; high when OUT > 10.76 V (R215 47 k / R216 5.9 k)	FLT → FLT_6A; **low = fault**. SPLYGD not used
EN net	EFUSE_USB_EN	EFUSE_12V_EN	EFUSE_6A_EN
IN / OUT caps	C207 / C208, 1 µF 50 V 0805	C209 / C210	C211 / C212
Output clamp diode	D201 B340A	D202 B340A	D203 B340A

How the numbers come out:

- Divider thresholds: V_trip = 1.2 V × (R_top + R_bottom) / R_bottom. For example, 1.2 × (100 + 24.9) / 24.9 = 6.02 V.
- Soft-start: TPS259474 slew = 2000 / C(pF) V/ms; TPS259480 slew = 5000 / C(pF) V/ms.
- Each current limit sits 15–31% above its rail's rated current (3.76 vs 3 A, 1.31 vs 1 A, 6.93 vs 6 A). Normal loads never trip it, but a short does.
- All three over-voltage cutoffs are well under the parts' 23 V maximum.

The clamp diodes (cathode to OUT, anode to GND) catch the negative kick when a long or inductive cable is suddenly cut off. PWR_FLAG symbols on 5V_BUS and 12V_BUS tell KiCad these nets are driven from another sheet.

The U201–U203 footprints are left blank on purpose. Import TI's official footprints from Ultra Librarian, because a hand-drawn QFN or WCSP land pattern is an easy way to ruin a board.

MCU sheet (page 4)

U301 is a Raspberry Pi Pico: SC0917 (Pico H, headers fitted) plugged into 2×20 female headers, or an SC0915 soldered down. Its footprint is the custom Thor_Power:RaspberryPi_Pico_TH (2.54 mm pitch, rows 17.78 mm apart).

Pico pin	GPIO	Net	Dir	What it does
4	GP2	BMS_SDA	Bidir	I2C1 data to the BMS (battery level, battery temp)
5	GP3	BMS_SCL	Out	I2C1 clock to the BMS
6	GP4	I2C_SDA	Bidir	I2C0 data to both INA228s
7	GP5	I2C_SCL	Out	I2C0 clock to both INA228s
9	GP6	ALERT_5V	In	Low = the 5V monitor flagged over-current or under-voltage
10	GP7	ALERT_12V	In	Low = the 12V monitor flagged
11	GP8	EFUSE_USB_EN	Out	High = USB output on, low = off or reset
12	GP9	EFUSE_12V_EN	Out	High = 12V output on
14	GP10	EFUSE_6A_EN	Out	High = 5V 6A output on
15	GP11	PG_USB	In	High = USB output voltage is good
16	GP12	PG_12V	In	High = 12V output voltage is good
17	GP13	FLT_6A	In	**Low** = the 6A eFuse is in fault
19	GP14	REG_5V_EN	Out	5V buck enable
20	GP15	REG_12V_EN	Out	12V regulator enable

Power pins: pin 39 VSYS is fed from 5V_BUS through D204 (B340A Schottky). That gives about 4.6 V, inside VSYS's 1.8–5.5 V range, and lets a laptop USB cable stay plugged in at the same time. Pin 36 3V3(OUT) supplies the board's +3V3 net. All 8 ground pins (3, 8, 13, 18, 23, 28, 33 AGND, 38) go to GND.

Marked no-connect and free for later: GP0/GP1 (kept for a UART debug port), GP16–GP22, GP26–GP28 (ADC-capable, for example a battery thermistor), RUN, ADC_VREF, 3V3_EN, and VBUS.

Pull-ups, all to +3V3:

Refs	Value	Nets	Why
R204, R205, R206	**10 k** (was 100 k)	EFUSE_USB_EN, EFUSE_12V_EN, EFUSE_6A_EN	Outputs default ON if firmware hangs; 10 k beats the RP2040's 50–80 k boot pull-down
R217, R218, R219	10 k	PG_USB, PG_12V, FLT_6A	PG and FLT are open drain and need a pull-up

Firmware should switch off the RP2040's internal pull-downs on GP6–GP13 at boot. That lets the external pull-ups set clean logic levels. The sheet notes now say this too.

Verification

There were three passes. Each one checks something different.

#	Check	How	Result
1	Wiring	Exported the netlist from KiCad 7 and compared every pin against a hand-written list of what each net should connect to	42/42 nets match, 17/17 no-connects intentional, 192 pins accounted for
2	Values vs datasheet math	Recomputed every current limit, divider threshold, slew rate, shunt voltage and power, and register value from the formulas above, then compared them to the values on the schematic	All match. Found the EN pull-up issue (fixed to 10 k)

#	Check	How	Result
3	Team design + BOM	Compared against the block diagram, the agreed parts, and the BOM reference by reference	2 monitors, 3 eFuses and the RP2040 match what was agreed. D204, the Pico and the 10 k change were missing from the BOM and are now added

Logic levels were also checked. Every signal that reaches a Pico pin is pulled up to 3.3 V, never 5 V or 12 V, so nothing over-drives the RP2040. The BMS lines are the one exception still open, because their pull-ups live on the BMS sheet.

ERC result (run in KiCad, Oct 25): 0 errors, 6 warnings.

Warning	Meaning	Action
U101, U102 footprint VSSOP-10_3x3mm_P0.5mm not found	That name isn't in this KiCad version's Package_SO library	Change both to Package_SO:MSOP-10_3x3mm_P0.5mm, which has the same 3×3 mm, 0.5 mm-pitch land pattern as TI's DGS package
BMS_SDA, BMS_SCL, REG_5V_EN, REG_12V_EN connected to one pin	These nets end at teammates' sheets, which aren't merged yet	Expected. They clear once the BMS and regulator sheets are joined

The 4 ignored tests are KiCad's defaults. After merging, turn on "Global label only appears once". It catches any interface label that's misspelled on one side.

Merge checklist and open items

Before joining the sheets:

- [ ] Teammates' sheets use exactly these names: 5V_REG, 12V_REG, 5V_USB_OUT, 5V_6A_OUT, 12V_1A_OUT, REG_5V_EN, REG_12V_EN, BMS_SDA, BMS_SCL, GND
- [ ] No other sheet has a +3V3 power symbol, or the team agrees the Pico is the only 3.3 V source

- [ ] **The 5V buck's EN defaults ON by itself.** The Pico is powered by the 5V buck, so the buck must start before the Pico can do anything. Its EN pull-up needs to be strong enough (≤ 10 k) to beat the RP2040's boot pull-down
- [ ] The regulator EN pull-ups must not go to battery or 48 V where they meet a Pico pin. The RP2040 tolerates 3.3 V at most. If a regulator's EN needs to sit at a high voltage, put an NPN or N-MOSFET on the regulator sheet and let the Pico drive its gate
- [ ] BMS sheet adds SMBus pull-ups to **3.3 V** (not 5 V) on BMS_SDA and BMS_SCL
- [ ] Swap the U101/U102 footprints to MSOP-10_3x3mm_P0.5mm, then re-run ERC after merging
- [ ] Import TI footprints for U201–U203 from Ultra Librarian

Decisions the team should confirm:

- **Where the 5V monitor sits.** In the schematic, U101 measures the whole 5V bus (USB and 6A outputs together). The earlier plan said "one for the 5V 6A rail". To monitor only the 6A output, move RS101 between the U203 split and U203's input. That changes nothing else.
- **Without the Pico, the board is dark.** +3V3 comes from the Pico, so if the Pico is unplugged, the eFuse EN pull-ups, INA228s and PG pull-ups all lose power and every output stays off. Plugged in but with hung firmware, the outputs still default ON. If outputs must work with no MCU at all, the board needs its own small 3.3 V LDO.
- **Cold rating.** The RP2040 is rated −20 to 85 °C, and a balloon can see -40 °C or colder. The eFuses and INA228s are rated -40 °C. Plan on insulation or a heater, or test the Pico cold.
- **Assembly.** The TPS259480AYWPR (WCSP-12 balls) needs a stencil and reflow, and the QFN-10 parts need hot air or reflow. Hand-soldering isn't practical. Decide custom PCB vs perfboard before ordering.
- **Old doc cleanup.** The earlier "Thor Redesign: Power Monitoring Circuit" doc still shows the old shunt number (WSL2512R0050FEA) and says VS = +3V3 without naming the Pico as the source.

Sources

- INA228 datasheet (TI)
- TPS25947 datasheet (TI)
- TPS25948 datasheet (TI)
- Raspberry Pi Pico datasheet

- RP2040/RP2350 GPIO pull-downs at reset (EEVblog)
- Files: Thor_Power_System_KiCad.zip, Thor_Power_System_schematic.pdf, Thor_Monitoring_Protection_BOM.csv